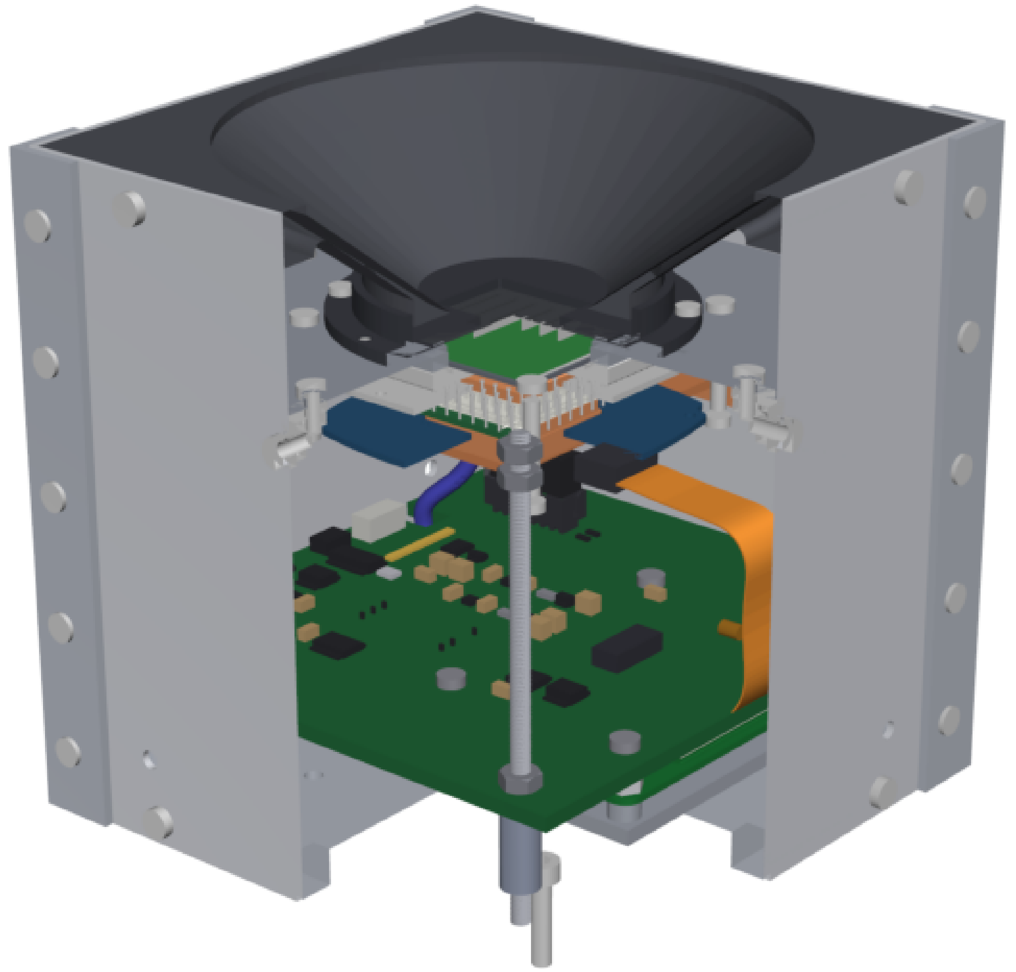
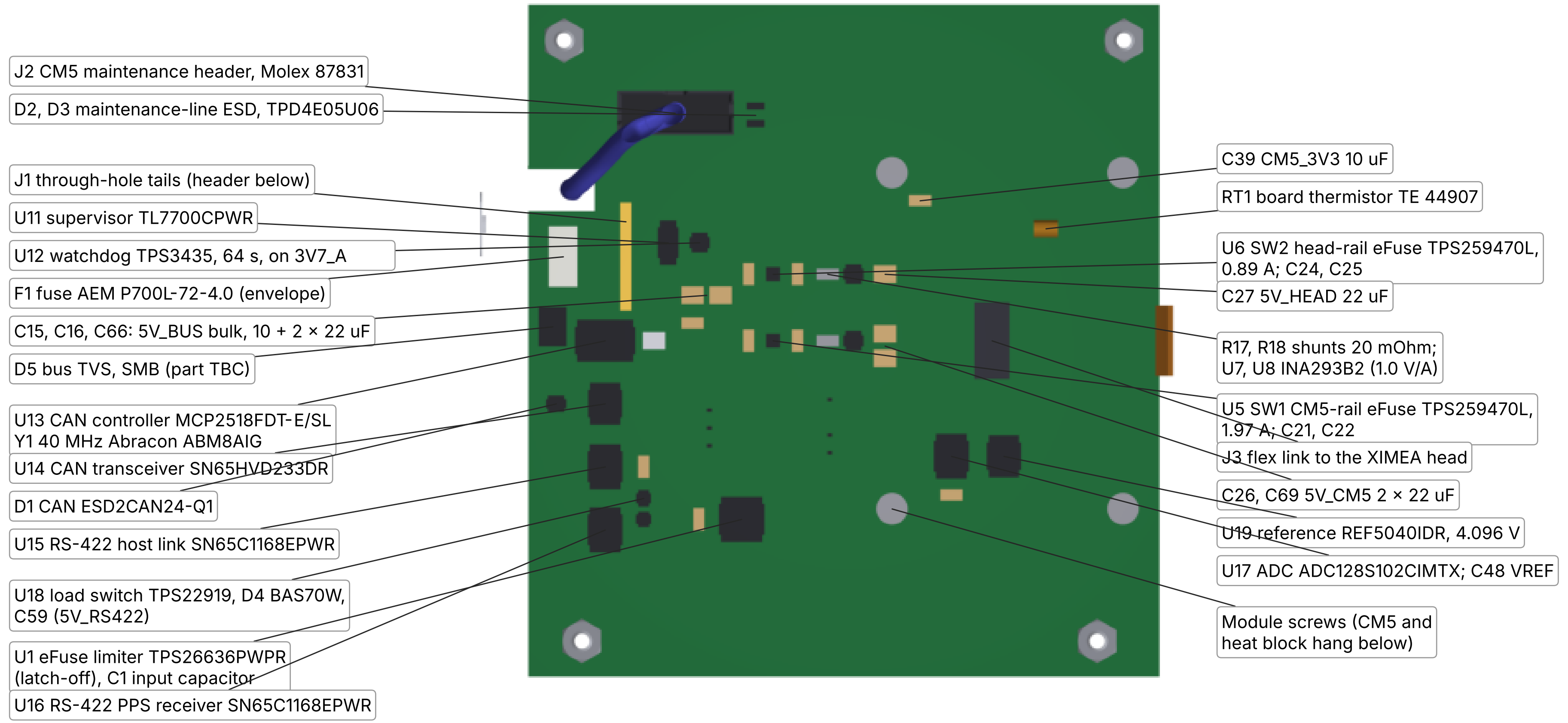
SLANT 1U carrier board in the housing: commercial parts, eFuse limiter, REF5040 reference, Gecko J1 and ceramic input filter

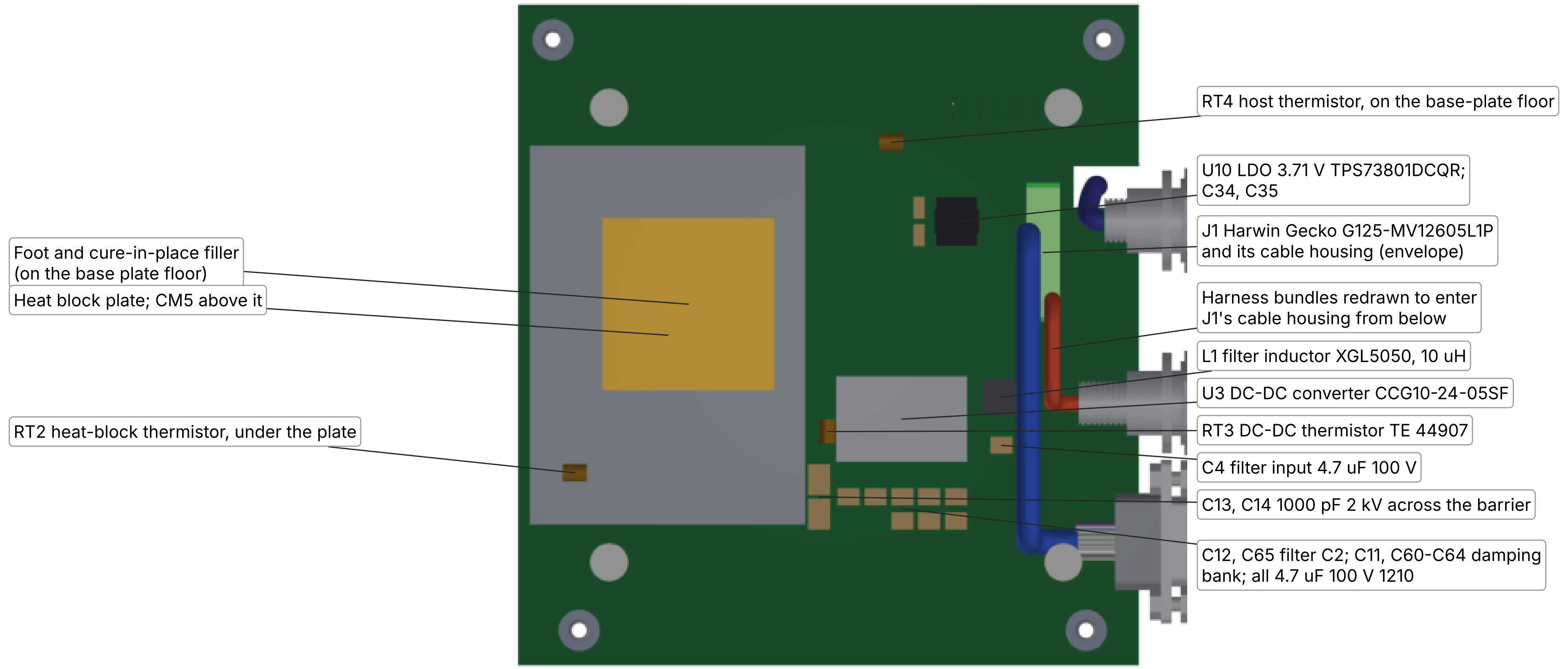
A. Cutaway, near +x/-y quadrant of the housing removed



B. Board from above (top side). References as in the schematic;


J2 CM5 maintenance header, Molex 87831
D2, D3 maintenance-line ESD, TPD4E05U06
J1 through-hole tails (header below)
U11 supervisor TL7700CPWR
U12 watchdog TPS3435, 64 s, on 3V7_A
F1 fuse AEM P700L-72-4.0 (envelope)
C15, C16, C66: 5V_BUS bulk, 10 + 2 × 22 uF
D5 bus TVS, SMB (part TBC)
U13 CAN controller MCP2518FDT-E/SL
Y1 40 MHz Abracon ABM8AIG
U14 CAN transceiver SN65HVD233DR
D1 CAN ESD2CAN24-Q1
U15 RS-422 host link SN65C1168EPWR
U18 load switch TPS22919, D4 BAS70W, C59 (5V_RS422)
U1 eFuse limiter TPS26636PWPR (latch-off), C1 input capacitor
U16 RS-422 PPS receiver SN65C1168EPWR
C39 CM5_3V3 10 uF
RT1 board thermistor TE 44907
U6 SW2 head-rail eFuse TPS259470L, 0.89 A; C24, C25
C27 5V_HEAD 22 uF
R17, R18 shunts 20 mOhm; U7, U8 INA293B2 (1.0 V/A)
U5 SW1 CM5-rail eFuse TPS259470L, 1.97 A; C21, C22
J3 flex link to the XIMEA head
C26, C69 5V_CM5 2 × 22 uF
U19 reference REF5040IDR, 4.096 V
U17 ADC ADC128S102CIMTX; C48 VREF
Module screws (CM5 and heat block hang below)

C. Board from below (x reversed). The CM5 is hidden by its heat block


Foot and cure-in-place filler (on the base plate floor)
Heat block plate; CM5 above it
RT2 heat-block thermistor, under the plate
RT4 host thermistor, on the base-plate floor
U10 LDO 3.71 V TPS73801DCQR; C34, C35
J1 Harwin Gecko G125-MV12605L1P and its cable housing (envelope)
Harness bundles redrawn to enter J1's cable housing from below
L1 filter inductor XGL5050, 10 uH
U3 DC-DC converter CCG10-24-05SF
RT3 DC-DC thermistor TE 44907
C4 filter input 4.7 uF 100 V
C13, C14 1000 pF 2 kV across the barrier
C12, C65 filter C2; C11, C60-C64 damping bank; all 4.7 uF 100 V 1210

Package sizes from the vendors' pages and drawings (TI, Coilcraft, Harwin, Abracon, AEM, Nexperia); heights are package maxima, assumed where the vendor gives none. Thermistor leads not drawn.